

Static Binding (Early Binding)

- **Definition:** The method or property call is resolved at compile time.
 - **How:** The compiler knows exactly which method to call based on the reference type.
 - **Keywords/Examples in C#:**
 - Normal method calls (non-virtual).
 - Method overloading.
 - **Performance:** Faster (no runtime lookup).
 - **Polymorphism:** Not supported — behavior is fixed at compile time.
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Dynamic Binding (Late Binding)

- **Definition:** The method or property call is resolved at runtime.
 - **How:** The actual object type (not the reference type) decides which method to execute.
 - **Keywords/Examples in C#:**
 - virtual, override, abstract methods.
 - dynamic keyword in C#.
 - **Performance:** Slightly slower (runtime lookup).
 - **Polymorphism:** Supported — enables runtime polymorphism.
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Quick Rule:

- **Static binding** → Chosen at compile time (overloading, non-virtual).
- **Dynamic binding** → Chosen at runtime (overriding, polymorphism).